Artificial Intelligence & Data Science

2-Mark Questions

Q.1 Write short note on Box Plot.

Q.2 Compare barplot and histogram.

Q.3 Differentiate Forward & Backward chaining.

Q.4 What is a histogram? Can we perform univariate graphical analysis using histogram.

Q.5 Explain various measures of the central tendencies of a statistical distribution.

Q.6 What are the different ways of knowledge representation.

Q.7 Compare Linear Regression vs. Logistic Regression.

Q.8 What is the difference between univariate, bivariate and Multivariate analysis.

Q.9 Differentiate correlation & covariance.

Q.10 What are different issues in ML algorithm.

Q.11 What do you mean by skolemization & unification.

Q.12 List out logical connectives used in AI.

5-Mark Questions

Q.1 Describe steps for developing ML applications with suitable diagram.

Q.2 Write short note on ANOVA / problem on ANOVA.

Q.3 Write short note on covariance & correlation / problem on the same.

Q.4 What are different univariate plots in EDA. Explain in detail.

Q.5 Compare z test, t test & ANOVA.

Q.6 What is linear regression? Explain its significance in ML and compare it with logistic regression.

Q.7 Sentence representation using FOL/predicate logic. Solve the problem using resolution.

Q.8 Write short note on planning techniques explain with example.

Q.9 What are different types of machine learning algorithms? Give example of each category.

Q.10 Define and state effects of overfitting and underfitting.

ANSWERS For 2-Mark Questions

Q1. Write short note on Box Plot.

- A **Box Plot** is a graphical representation of data distribution.
- It shows the **minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum** values.
- The "box" represents the **interquartile range (IQR = Q3 − Q1)**.
- **Whiskers** extend to minimum and maximum values, highlighting spread.
- Useful for detecting **outliers** and understanding data variability.

Q2. Compare barplot and histogram.

Feature	Bar Plot	Histogram
Data Type	Categorical data	Continuous numerical data
Bars	Separated with gaps	Touch each other (no gaps)
Purpose	Compare categories	Show frequency distribution
X-axis	Categories or discrete values	Intervals (bins) of continuous data

Feature	Bar Plot	Histogram
Use Case	Compare sales by product, votes by party	Analyze exam scores, age distribution

Q3. Differentiate Forward & Backward chaining.

Aspect	Forward Chaining	Backward Chaining
Direction	Data → Conclusion	Goal → Data
Process	Starts from known facts, applies rules to reach new facts	Starts from a goal, works backward to check supporting facts
Use Case	Suitable for **prediction & simulation**	Suitable for **diagnosis & problem-solving**
Example	From symptoms → predict disease	From suspected disease → verify symptoms

Q4. What is a histogram? Can we perform univariate graphical analysis using histogram

- A **Histogram** is a graphical representation of the **frequency distribution** of continuous numerical data.
- Data is divided into **intervals (bins)**, and the height of each bar shows how many values fall within that range.
- Bars are **adjacent (no gaps)** since the data is continuous.
- Yes, histograms are widely used for **univariate graphical analysis**, as they show the distribution of a **single variable** clearly.

Q5. Explain various measures of the central tendencies of a statistical distribution.

- **Mean (Average):** Sum of all values divided by the number of values.
- **Median:** Middle value when data is arranged in ascending order; less affected by outliers.
- **Mode:** Value that occurs most frequently in the dataset.

- **Midrange:** Average of the minimum and maximum values.

Q6. What are the different ways of knowledge representation.

- **Logical Representation:** Uses formal logic (propositional, predicate logic) to represent facts and rules.
- **Semantic Networks:** Represents knowledge as nodes (concepts) and edges (relationships).
- **Frames:** Data structures that hold attributes and values for representing stereotypical situations.
- **Ontologies:** Structured representation of concepts and relationships in a domain.

Q7. Compare Linear Regression vs. Logistic Regression.

Aspect	Linear Regression	Logistic Regression
Purpose	Predicts continuous values	Predicts categorical outcomes (binary/multi)
Output Range	Any real number	Between 0 and 1 (probability)
Model Type	Regression model	Classification model
Use Case	Predict house prices, sales	Predict disease presence, spam detection
Equation	$Y = \beta_0 + \beta_1 X$	$P(Y) = \frac{1}{1+e^{-(\beta_0+\beta_1 X)}}$

Q8. What is the difference between univariate, bivariate and Multivariate analysis.

Aspect	Univariate Analysis	Bivariate Analysis	Multivariate Analysis
Variables	Single variable	Two variables	More than two variables
Purpose	Describe distribution of one variable	Study relationship between two	Analyze interactions among multiple

Aspect	Univariate Analysis	Bivariate Analysis	Multivariate Analysis
		variables	variables
Techniques	Histogram, Box plot, Mean/Median	Scatter plot, Correlation, Regression	Multivariate regression, PCA, MANOVA
Example	Distribution of student marks	Marks vs Study hours	Marks vs Study hours vs Attendance

Q9. Differentiate correlation & covariance.

Aspect	Correlation	Covariance
Definition	Measures the **strength & direction** of relationship between variables	Measures the **degree to which two variables vary together**
Range	Always between **-1 and +1**	Can take **any value** (positive, negative, or zero)
Scale Dependence	Independent of units (standardized)	Depends on units of variables
Interpretation	Shows how strongly variables are related	Shows whether variables move in same or opposite direction

Q10. What are different issues in ML algorithm.

- **Overfitting:** Model learns noise in training data, performs poorly on unseen data.
- **Underfitting:** Model is too simple, fails to capture patterns in data.
- **Data Quality Issues:** Missing values, noisy or imbalanced datasets affect performance.
- **Computational Complexity:** High training time and resource requirements for large datasets.

Q11. What do you mean by skolemization & unification.

- **Skolemization:**

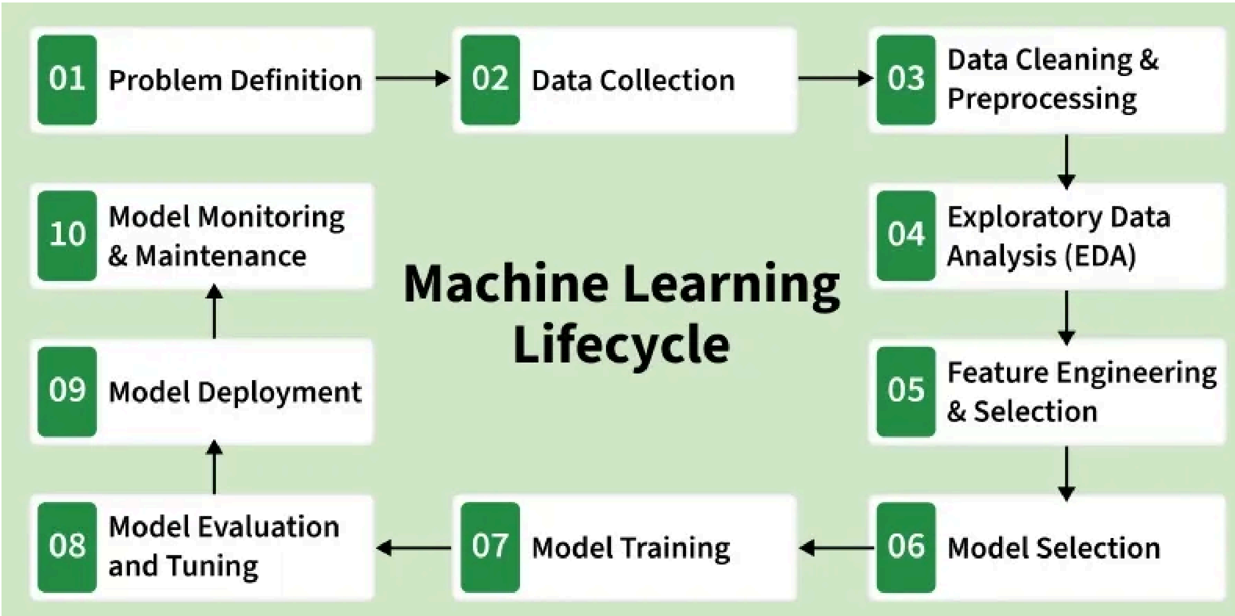
    - A process in **predicate logic** to remove existential quantifiers.
    - Replaces them with **Skolem functions or constants**.
    - Simplifies formulas for automated reasoning.
- **Unification:**
    - A technique in logic and AI to make two expressions **identical** by finding a suitable substitution of variables.
    - Used in **inference engines** and **Prolog programming**.
    - Enables rule matching and logical deduction.

Q12. List out logical connectives used in AI.

- **AND (∧)** → True only if both statements are true.
- **OR (∨)** → True if at least one statement is true.
- **NOT (¬)** → Negates the truth value of a statement.
- **IMPLICATION (→)** → If first statement is true, then second must be true.
- **BICONDITIONAL (↔)** → True when both statements have the same truth value.

ANSWERS For 5-Mark Questions

Q1. Describe steps for developing ML applications with suitable diagram.



Steps for Developing ML Applications

1. **Data Collection**
    - Gather raw data from sources (databases, sensors, files, APIs).
2. **Data Preprocessing**
    - Clean data (handle missing values, remove noise).
    - Transform and normalize features for consistency.
3. **Model Selection**
    - Choose suitable algorithm (e.g., regression, classification, clustering).
4. **Model Training**
    - Feed training data into the algorithm.
    - Adjust parameters to learn patterns.
5. **Model Evaluation**
    - Test model on validation/test data.
    - Use metrics like accuracy, precision, recall, RMSE.
6. **Deployment & Monitoring**

- Deploy model into production environment.
- Continuously monitor performance and update when needed.

Q2. Write short note on ANOVA / problem on ANOVA.

Definition:
ANOVA (Analysis of Variance) is a **statistical technique** used to compare the **means of three or more groups** to determine if there are significant differences among them.

Steps in ANOVA

1. **State Hypotheses:**
   - Null hypothesis (H_0): All group means are equal.
   - Alternative hypothesis (H_1): At least one group mean differs.
2. **Calculate Group Means and Overall Mean.**
3. **Compute Sum of Squares:**
   - **Between-group variance (SSB)** — variation due to differences between group means.
   - **Within-group variance (SSW)** — variation within each group.
4. **Calculate F-Ratio:**

$$F = \frac{\text{Mean Square Between (MSB)}}{\text{Mean Square Within (MSW)}}$$

5. **Compare F-value with Critical Value:**
   - If $F > F_{critical}$, reject H_0.
6. **Draw Conclusion:**
   - Conclude whether group means differ significantly.

Q3. Write short note on covariance & correlation / problem on the same.

Covariance

- Covariance measures how **two variables change together**.
- If both increase or decrease simultaneously → **positive covariance**.
- If one increases while the other decreases → **negative covariance**.
- Formula:

$$\mathrm{Cov}(X,Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1}$$

- It indicates the **direction** of the relationship but not its strength.
- Units depend on the variables, making comparison difficult.

* Problem on Co-variance :

X	Y
1	2
2	4
3	6
4	8
5	10

N = 5.

→ $\bar{X} = 3$ $\quad \bar{Y} = 6$ → Mean

formula of Co-variance

$$Cov(x,y) = \frac{1}{N} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{5}\Big[(1-3)(2-6) + (2-3)(4-6) + (3-3)(6-6) + (4-3)(8-6) + (5-3)(10-6)\Big]$$

$$= \frac{1}{5}\left[8 + 2 + 0 + 2 + 8\right]$$

$$= \frac{\cancel{20}^{4}}{\cancel{5}_{1}} = \boxed{4}$$

$Cov(x,y) = 4$

which is greater or equals to 0

that means it is a +ve Co-variance

Correlation

- Correlation measures the **strength and direction** of the linear relationship between two variables.
- It is a **standardized form of covariance**.
- Formula:

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Range: **$-1 \le r \le +1$**

- $r=+1$: Perfect positive correlation
- $r=-1$: Perfect negative correlation
- $r=0$: No linear relationship

* Problem on Pearson's Co-relation.

Formula To find the co-relation

$$r = \frac{Cov(x,y)}{\sigma x \sigma y}$$

We have evaluated the co-variance before now, we need to calculate std. devaition of x & y.

$(x_i-\bar{x})^2$	$(y_i-\bar{y})^2$
4	16
1	4
0	0
1	4
4	16
Total = 10	Total = 40

$Avg = \frac{10}{5} = \boxed{2}$ $\quad$ $Avg = \frac{40}{5} = \boxed{8}$

$$r = \frac{4}{\sqrt{2} \cdot \sqrt{8}} = \frac{4}{1.41 \times 2.82}$$

$r = \frac{4}{3.97}$ $\quad$ $\boxed{\therefore r = 1.00}$

Q4. What are different univariate plots in EDA. Explain in detail.

Definition:
Univariate plots are graphical techniques used to analyze the distribution and characteristics of a **single variable**. They help in understanding central tendency, spread, skewness, and presence of outliers.

Types of Univariate Plots

1. **Histogram**
   - Displays frequency distribution of continuous data.
   - Bars touch each other since data is continuous.
   - Useful for identifying skewness and spread.
2. **Box Plot (Whisker Plot)**
   - Summarizes data using **minimum, Q1, median, Q3, maximum**.
   - Highlights **outliers** clearly.
   - Useful for comparing distributions across categories.
3. **Density Plot**
   - Smooth curve showing probability distribution of a variable.
   - Useful for visualizing distribution shape more clearly than a histogram.
4. **Bar Plot**
   - Used for categorical variables.
   - Shows frequency or proportion of each category.
5. **Pie Chart (less common in EDA)**
   - Represents categorical data as proportions of a whole.
   - Less preferred due to difficulty in comparing slices.

Example Use Cases

- Histogram → Distribution of student marks.
- Box Plot → Detecting salary outliers in a company dataset.
- Density Plot → Understanding distribution of customer ages.
- Bar Plot → Frequency of product sales categories.

Q.5 Compare z test, t test & ANOVA.

Aspect	Z-Test	T-Test	ANOVA (Analysis of Variance)
Purpose	Tests whether sample mean differs from population mean (large samples)	Tests whether means of two groups differ (small samples)	Tests whether means of **3 or more groups** differ
Sample Size	Large sample size (n>30)	Small sample size (n<30)	Any sample size, but usually multiple groups
Population Variance	Assumed known	Assumed unknown	Variance within and between groups compared
Distribution Used	Standard Normal Distribution (Z)	Student's t-distribution	F-distribution
Hypotheses	H0: Sample mean = Population mean	H0: Means of two groups are equal	H0: All group means are equal
Use Case	Quality control, large-scale testing	Comparing exam scores of two classes	Comparing effectiveness of multiple teaching methods

Q.6 What is linear regression? Explain its significance in ML and compare it with logistic regression.

Linear Regression

Definition:

Linear Regression is a **supervised learning algorithm** used to model the relationship between a **dependent variable (Y)** and one or more **independent variables (X)** by fitting a straight line.

Equation: $Y = \beta 0 + \beta 1 X + \epsilon$

Where:

- Y = dependent variable
- X = independent variable
- $\beta 0$ = intercept
- $\beta 1$ = slope (coefficient)
- ϵ = error term

Significance in ML

1. **Predictive Modeling:** Used for forecasting continuous outcomes (e.g., sales, prices).
2. **Interpretability:** Coefficients show how much change in X affects Y.
3. **Baseline Model:** Often the first step before applying complex ML algorithms.
4. **Feature Importance:** Helps identify which variables strongly influence the target.
5. **Simplicity & Efficiency:** Easy to implement and computationally inexpensive.

Comparison: Linear vs Logistic Regression

Aspect	Linear Regression	Logistic Regression
Purpose	Predicts continuous values	Predicts categorical outcomes (binary/multi)
Output Range	Any real number	Between 0 and 1 (probability)
Model Type	Regression model	Classification model
Use Case	Predict house prices, sales	Predict disease presence, spam detection

Equation	$Y = \beta_0 + \beta_1 X$	$P(Y) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$

Q.8 Write short note on planning techniques explain with example.

Definition:
Planning in AI refers to the process of generating a sequence of actions that an agent must perform to achieve a specific goal. It is a core component of problem-solving and decision-making systems.

Planning Techniques

1. **State-Space Representation**
   - Represents problems as states and transitions.
   - Example: Pathfinding in a maze where each position is a state.
2. **Forward Planning (Progression Search)**
   - Starts from the initial state and applies actions step by step until the goal is reached.
   - Example: Robot moving from start to destination.
3. **Backward Planning (Regression Search)**
   - Starts from the goal state and works backward to the initial state.
   - Example: Planning steps needed to solve a puzzle by imagining the final arrangement first.
4. **Hierarchical Planning**
   - Breaks down complex goals into sub-goals.
   - Example: Planning a trip → Book tickets, arrange accommodation, plan sightseeing.
5. **Partial-Order Planning**
   - Plans actions without fixing their exact order initially, allowing flexibility.

- Example: Cooking dinner → chopping vegetables and boiling water can be parallel tasks.

6. **Conditional Planning**
   - Considers uncertainty and alternative paths depending on conditions.
   - Example: Weather-based planning → "If it rains, carry an umbrella; else, wear sunglasses."

Example:

Suppose an AI agent is planning to **deliver a package**:

- **State-space:** Locations of package and destination.
- **Forward planning:** Start at warehouse → load package → drive → deliver.
- **Conditional planning:** If traffic jam → take alternate route.

Q.9 What are different types of machine learning algorithms? Give example of each category.

Machine Learning algorithms can be broadly classified into **three main categories** based on the type of task and supervision involved.

1. Supervised Learning

- **Definition:** Learns from labeled data (input–output pairs).
- **Goal:** Predict outcomes for new unseen data.
- **Examples:**
  - **Linear Regression** → Predict house prices.
  - **Logistic Regression** → Spam email detection.
  - **Decision Trees / Random Forests** → Classify customer churn.
  - **Support Vector Machines (SVM)** → Handwritten digit recognition.

2. Unsupervised Learning

- **Definition:** Learns from unlabeled data, finds hidden patterns.
- **Goal:** Group or reduce data without predefined labels.
- **Examples:**
  - **Clustering (K-Means, Hierarchical)** → Customer segmentation.
  - **Principal Component Analysis (PCA)** → Dimensionality reduction in image data.
  - **Association Rule Learning (Apriori, FP-Growth)** → Market basket analysis.

3. **Reinforcement Learning**

- **Definition:** Agent learns by interacting with environment, receiving rewards/penalties.
- **Goal:** Maximize cumulative reward through trial and error.
- **Examples:**
  - **Q-Learning** → Robot navigation.
  - **Deep Reinforcement Learning (DQN)** → Playing video games like Atari.
  - **Policy Gradient Methods** → Self-driving cars decision-making.

Q.10 Define and state effects of overfitting and underfitting.

Definition

- **Overfitting:**
  A model learns the training data **too well**, including noise and random fluctuations, leading to poor generalization on unseen data.
- **Underfitting:**
  A model is **too simple** to capture the underlying patterns in data, resulting in poor performance on both training and test sets.

Effects of Overfitting

1. High accuracy on training data but low accuracy on test data.
2. Model becomes complex and memorizes noise.
3. Poor generalization to new/unseen datasets.
4. Leads to misleadingly optimistic training results.
5. Requires techniques like regularization, pruning, or more data to fix.

Effects of Underfitting

1. Low accuracy on both training and test data.
2. Model fails to capture important relationships.
3. Predictions are overly simplistic.
4. Bias is high, variance is low.
5. Can be improved by using more complex models or better features.